Comparison of the reference and agent-prepared datasets

Shanjie Jiao

Assessment and comparison method

`final_data.csv` is the accepted dataset for this assignment. It is generated by `Script/data_script.r`. The alternative `final_data_replication.csv` is generated independently by ChatGPT in `Script/research_replication.R`. Both scripts read the seven raw datasets directly; neither requires the other script's output or a saved reference snapshot.

The datasets differ in **13 `armed_conflict` values**: the accepted dataset assigns 1 and the replication assigns 0. The accepted dataset has **704** exposed country-years, compared with **691** in the replication. The other 16 variables and missing-value positions match.

Rows are aligned by `iso` and `year` before every column is compared. Paired missing values count as agreement; a missing value paired with an observed value counts as a difference. Duplicate keys and different country-year coverage cause validation to fail. Rendering this document only reads the two final datasets and raw conflict data.

Structure and variable-level differences

Table 1: Structure and conflict exposure counts for each dataset.

Dataset	Rows	Columns	Countries	Years	Exposed
Accepted_final	3720	17	186	2000–2019	704
Agent_replication	3720	17	186	2000–2019	691

Both datasets have 3,720 rows and 17 variables, covering 186 countries during 2000–2019. There are no duplicate country-year keys or unmatched rows.

Table 2: Number of differing cells relative to the accepted reference.

Variable	Differing cells
iso	0
year	0
armed_conflict	13
maternal_mortality	0
infant_mortality	0
under5_mortality	0
neonatal_mortality	0
drought	0
earthquake	0
gdp_1000	0
oecd	0
pop_dens	0
urban	0
age_dep	0
male_edu	0
temp	0
rainfall_1000	0

The replication differs from the accepted dataset in 13 cells. The 13 affected country-years represent approximately 0.35% of the panel.

What the agent changed and why the outputs differed

The accepted original calculation sums `best` across **all events in each country-year** and sets `armed_conflict = 1` when that total is at least 25. It then shifts the year forward by one, so exposure in 1999 is associated with the outcome in 2000.

The agent previously changed `data_script.r` to group by **`iso, year, and conflict_id`** first. It applied the 25-death threshold to each individual conflict, then set the country-year indicator to 1 only when at least one conflict qualified. This was the rule already used in `research_replication.R`. Consequently, several separate conflicts with fewer than 25 deaths each could sum to 25 or more deaths, yet the agent's rule assigned 0.

The agent also added special handling for missing death estimates. The raw data contain no missing `best` values attached to nonmissing conflict IDs, so that change did not cause the 13 discrepancies. The one-year exposure shift, mortality preparation, disaster aggregation, covariate joins, and column order were retained.

The earlier report justified the change using the agent's interpretation of the paper. That interpretation did not establish that the assignment's accepted country-year rule should be replaced.

Agreement with another agent-generated output is not sufficient evidence that the original data are wrong. This report therefore evaluates the replication against the accepted `final_data.csv`.

Table 3: All 13 differences between the accepted dataset and the replication. Year refers to the outcome year.

ISO	Year	Exposure year	Reference	Agent	Total deaths	Largest conflict
BGD	2019	2018	1	0	28	23
BRA	2008	2007	1	0	40	21
BRA	2017	2016	1	0	34	12
COG	2004	2003	1	0	33	22
HND	2004	2003	1	0	47	24
KEN	2003	2002	1	0	45	22
PAK	2002	2001	1	0	42	20
SEN	2000	1999	1	0	31	23
TCD	2017	2016	1	0	26	18
THA	2002	2001	1	0	25	23
TUN	2019	2018	1	0	25	24
UGA	2009	2008	1	0	32	13
VEN	2003	2002	1	0	27	20

For example, Bangladesh's 2018 events total 28 deaths, but the largest individual conflict has 23. The reference correctly assigns 1 to Bangladesh's 2019 outcome row under the accepted country-year rule. The agent changed this to 0 by using the individual-conflict threshold. Restoration returns it to 1. The same explanation applies to every row in the table.

What was restored

The conflict section in `Script/data_script.r` was returned to the following calculation, and the script was rerun to generate `final_data.csv`:

```
confl <- confl0 |>
  group_by(iso, year) |>
  summarise(deaths = sum(best, na.rm = TRUE), .groups = "drop") |>
  transmute(
    iso,
    year = as.integer(year) + 1L,
    armed_conflict = as.integer(deaths >= 25)
  ) |>
  filter(between(year, 2000L, 2019L))
```

This reverses the 13 agent changes from 0 back to 1 and restores the total of 704 exposed country-years. No raw data were edited. `research_replication.R` and its output retain their alternative

definition so the discrepancy remains visible and explainable; they should not replace the accepted dataset.

Missingness and final validation

Table 4: Missing values by variable in the two final datasets.

Variable	Reference_NA	Replication_NA
iso	0	0
year	0	0
armed_conflict	0	0
maternal_mortality	426	426
infant_mortality	20	20
under5_mortality	20	20
neonatal_mortality	20	20
drought	0	0
earthquake	0	0
gdp_1000	62	62
oecd	0	0
pop_dens	20	20
urban	20	20
age_dep	0	0
male_edu	20	20
temp	20	20
rainfall_1000	20	20

Missing mortality and covariate values remain missing. They were not imputed or replaced with zero during restoration. In particular, maternal mortality has 426 missing values and GDP has 62. Matching the reference does not remove the need to address missingness in subsequent analyses.

Table 5: Validation of the accepted dataset and comparison results.

Check	Passed
Required 17 columns in both datasets	TRUE
Unique and matching country-year keys	TRUE
Accepted exposures match country-year totals from raw events	TRUE
704 accepted and 691 replication exposed country-years	TRUE
Other 16 variables match exactly	TRUE

The difference is caused by the aggregation level used for the conflict threshold, not by a merge error. The original country-year rule was restored in `data_script.r` after the agent had temporarily

replaced it with the replication's conflict-level rule. The earlier claim that matching the replication corrected the original data was withdrawn. The accepted output remains `final_data.csv`, and the alternative remains available for comparison.

Reproducing the two datasets separately

From the project root, run either command independently:

```
Rscript Script/data_script.r
Rscript Script/research_replication.R
```

The first command writes `Data/Processed_Data/final_data.csv`; the second writes `Data/Processed_Data/final_data_replication.csv`. No extra snapshot datasets are required. After generating both files, this report can be rendered or its R chunks run to repeat the comparison.